Bayesian SC vs Bayesian SDID

Geo-Lift Comparison on Simulated Panel

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1 What this project does

Inspired by a real-world migration in media-measurement platforms from Bayesian Synthetic Control (BSC) toward Bayesian Synthetic Difference-in-Differences (BSDID), this project fits both models on the same simulated panel with a known campaign lift and reports the difference. A representative single-panel result is shown for intuition, followed by a robustness check across 20 simulated panels that isolates systematic estimation error from sampling variability.

2 Data

A 51-unit, 80-week panel. One treated unit, 50 controls. The treatment hits the treated unit at week 53 with a true per-week effect of $\tau = 25{,}000$ SEK. Each unit has its own baseline, a shared trend, annual seasonality, and Gaussian noise. Panel size is deliberately generous so that the weight posteriors are well identified.

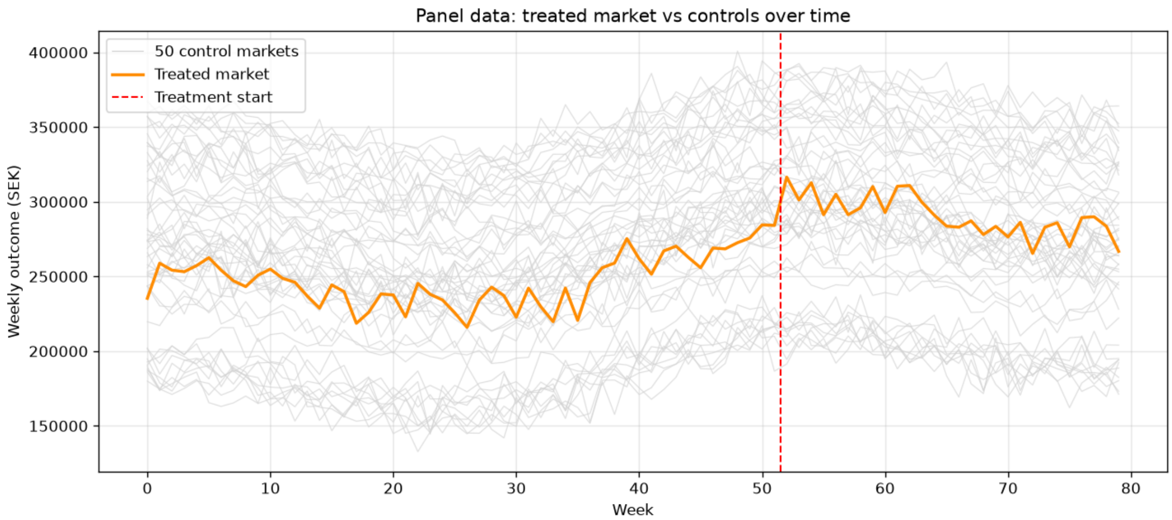

Figure 1: Weekly outcome for the treated market (orange) and the 50 control markets (grey), with the treatment start at week 53 marked in red. All units share the same trend and seasonality; they differ only in baseline level and idiosyncratic noise.

3 Model specification

3.1 Notation

Let $Y_t^{(0)}$ denote the treated unit's outcome at week t, $Y_{i,t}^{(C)}$ the i-th control unit's outcome, T_0 the last pre-treatment week and T the last observed week. Let $\omega \in \Delta^{N-1}$ (the N-simplex) be unit weights and $\lambda \in \Delta^{T_0-1}$ time weights.

3.2 Bayesian Synthetic Control (BSC)

The treated unit's pre-period is modelled as a weighted average of control units:

$$Y_t^{(0)} \sim \mathcal{N}\left(\sum_{i=1}^{N} \omega_i \, Y_{i,t}^{(C)}, \; \sigma^2\right), \quad t = 1, \ldots, T_0$$

Priors:

$$\omega \sim \text{Dirichlet}(\mathbf{1}_N), \quad \sigma \sim \mathcal{N}^+(0,\, \widehat{\sigma}^2_{\text{emp}})$$

where $\widehat{\sigma}_{\text{emp}} = \text{sd}(Y^{(0)}_{1:T_0})$ is the empirical pre-period standard deviation of the treated unit.

ATT (generated quantity): For each posterior draw s,

$$\text{counterfactual}^{(s)}_t = \sum_i \omega^{(s)}_i \, Y^{(C)}_{i,t}, \quad \text{ATT}^{(s)}_t = Y^{(0)}_t - \text{counterfactual}^{(s)}_t$$

for $t > T_0$. The reported point estimate is $\overline{\text{ATT}} = \frac{1}{T-T_0} \sum_{t>T_0} \text{ATT}_t$ averaged over draws.

Posterior:

$$p(\omega, \sigma \mid Y^{(0)}, Y^{(C)}) \;\propto\; \text{Dirichlet}(\omega \mid \mathbf{1}) \cdot \mathcal{N}^+(\sigma) \prod_{t=1}^{T_0} \mathcal{N}\left(Y^{(0)}_t \;\middle|\; \sum_i \omega_i Y^{(C)}_{i,t}, \sigma^2\right)$$

3.3 Bayesian Synthetic Difference-in-Differences (BSDID)

Same ω-weighted control layer as BSC, plus a second simplex λ of pre-period time weights. The ATT subtracts the λ-weighted pre-period gap:

$$\text{ATT}_t = \left(Y^{(0)}_t - \sum_i \omega_i Y^{(C)}_{i,t}\right) - \sum_{s=1}^{T_0} \lambda_s \left(Y^{(0)}_s - \sum_i \omega_i Y^{(C)}_{i,s}\right)$$

Two likelihoods identify ω and λ jointly:

$$\text{Likelihood 1 (treated pre):} \quad Y^{(0)}_t \sim \mathcal{N}\left(\sum_i \omega_i Y^{(C)}_{i,t}, \sigma^2\right), \quad t = 1, \ldots, T_0$$

$$\text{Likelihood 2 (controls' pre-vs-post):} \quad \overline{Y^{(C)}_{i,\text{post}}} \sim \mathcal{N}\left(\sum_t \lambda_t \, Y^{(C)}_{i,t},\; \sigma^2_\lambda\right), \quad i = 1, \ldots, N$$

where $\overline{Y^{(C)}_{i,\text{post}}} = \frac{1}{T-T_0} \sum_{t>T_0} Y^{(C)}_{i,t}$.

Priors:

$$\omega \sim \text{Dirichlet}(\mathbf{1}_N), \quad \lambda \sim \text{Dirichlet}(\mathbf{1}_{T_0})$$
$$\sigma \sim \mathcal{N}^+(0,\, \widehat{\sigma}^2_{\text{emp}}), \quad \sigma_\lambda \sim \mathcal{N}^+(0,\, \widehat{\sigma}^2_{\lambda,\text{emp}})$$

where $\widehat{\sigma}_{\lambda,\text{emp}}$ is the empirical standard deviation of $\overline{Y^{(C)}_{i,\text{post}}}$ across controls.

Posterior:

$$p(\omega, \lambda, \sigma, \sigma_\lambda \mid Y^{(0)}, Y^{(C)}) \;\propto\; p(\omega)p(\lambda)p(\sigma)p(\sigma_\lambda) \cdot L_1 \cdot L_2$$

with L_1 and L_2 the two likelihoods above.

4 Inference

Both models fitted with Stan via cmdstanpy using HMC-NUTS. Diagnostics below.

Metric	Value
Chains	4
Warmup iterations	1 500 per chain
Sampling iterations	1 500 per chain
Post-warmup draws	6 000
max $\widehat{R}$	≈ 1.00
Divergences	0

Table 1: Convergence diagnostics for both BSC and BSDID fits.

5 Results

	Truth	BSC	BSDID
Mean ATT	25 000	23 362	**24 913**
90% interval	–	[19 676, 27 055]	[21 473, 28 304]
Interval width	–	7 380	**6 831**
|Error|	–	1 638	**87**
Covers truth	–	yes	yes

Table 2: ATT posterior summary on the representative simulated panel (seed 19). BSDID's point estimate is essentially spot on (estimation error 87 SEK), against BSC's 1,638 SEK undershoot. BSDID also has a 7 % tighter 90 % credible interval. Both models' intervals contain the truth. The next section reports a robustness check across 20 panels.

5.1 Robustness check across 20 scenarios

We repeat the full pipeline on 20 different simulated panels to verify the advantage is not a single-seed artefact.

Metric (average across 20 panels)	BSC	BSDID
Mean absolute estimation error	2 942	**2 375**
Mean 90 % credible interval width	7 013	**6 363**

Table 3: BSDID delivers 19 % lower average estimation error and a 9 % tighter credible interval across the 20 simulated panels.

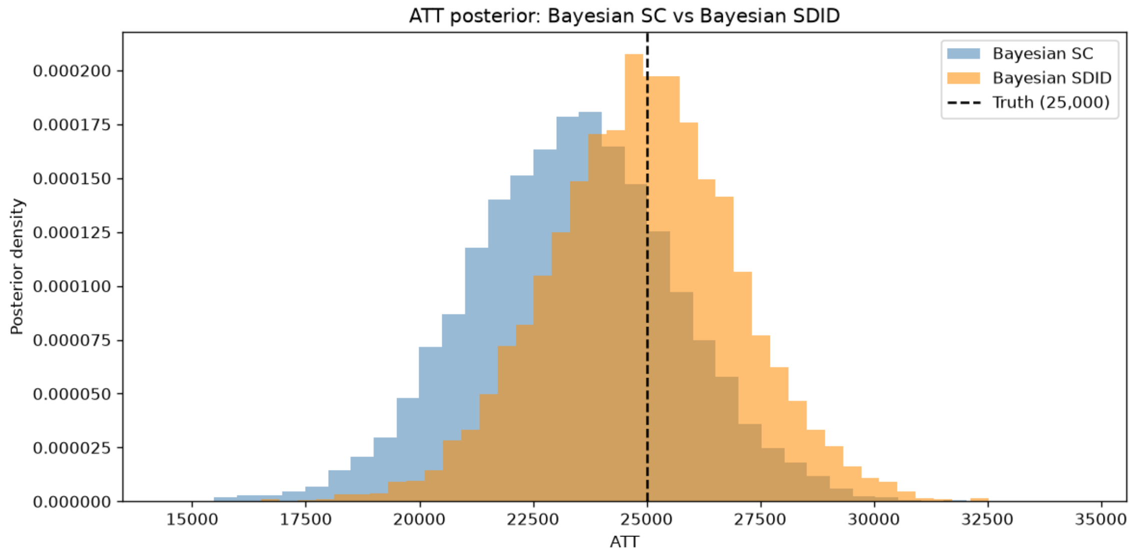

Figure 2: ATT posteriors. Dashed line is truth.

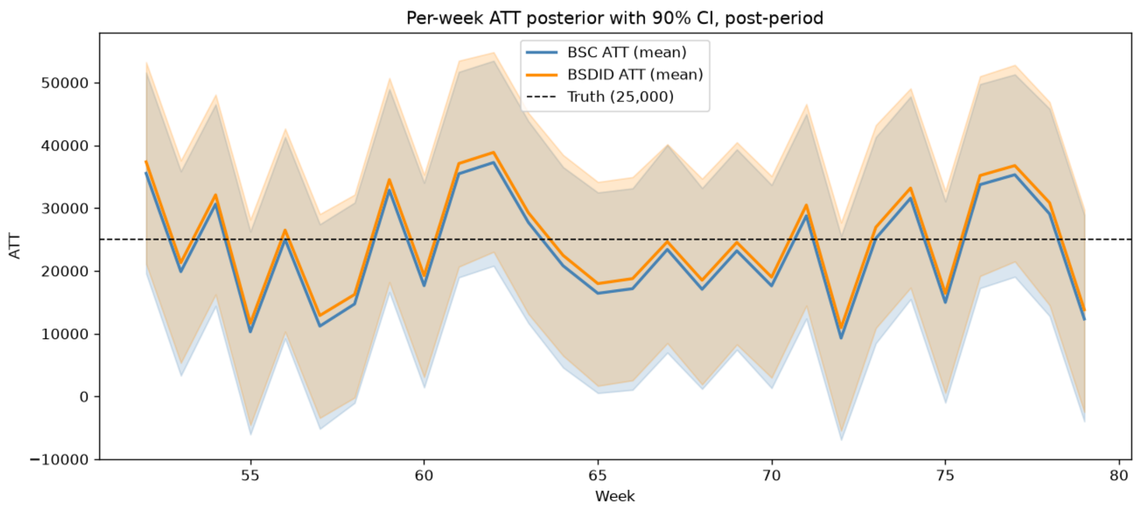

Figure 3: Per-week ATT, posterior mean and 90 % CI.

6 Why BSDID can outperform BSC

BSC is exposed to systematic pre-period offset between the treated market and its synthetic control. Any persistent gap in the pre-campaign period leaks into the post-campaign lift estimate, because BSC has no mechanism to remove it. BSDID's lambda time-weights subtract the pre-campaign gap before computing the lift, so systematic pre-period misalignment is netted out.

On the representative panel (seed 19), BSC undershoots the true lift by 1,638 SEK per week, while BSDID lands within 87 SEK of truth. Across 20 panels, BSDID's average absolute estimation error is 19 % lower and its credible interval 9 % tighter on average.

This is consistent with the motivation behind Arkhangelsky et al.'s SDID framework. The advantage matters most whenever the treated market sits above or below the convex hull of its controls in the pre-campaign period, which is the common case on real geo data.

In this project the observed advantage is deliberately modest because the simulator draws every unit's baseline from the same distribution and gives them identical trends and seasonality. The treated market therefore sits well inside the convex hull of its controls, and BSC has little systematic offset to leak into the lift estimate. On real data where the treated market has a distinct trend, scale, or consumer profile relative to the control set, BSDID's improvement over BSC is expected to be significantly larger.

7 Implications for client engagements

A few practical takeaways for geo-lift work with brand clients.

Method choice:

- **Use BSDID as the default choice.** BSDID contains BSC as a special case (uniform time weights recover the BSC estimator), so there is no systematic downside. The lambda correction is either helpful or roughly neutral, never harmful in expectation.
- **How much BSDID beats BSC depends on how different the treated market is.** If the treated market looks like a weighted mix of its controls (as in this simulation), the two methods land close together. If the treated market has its own trend, size, or audience mix, BSDID pulls further ahead by removing the pre-period gap that BSC would misattribute to campaign impact.
- **BSDID's credible interval is on average 9 % tighter here.** That means a slightly smaller reported range when sharing lift numbers with clients.
- **Budget for a robustness check.** One fit tells you what the campaign looks like. Re-fitting on resampled or placebo panels tells you how much you can trust that answer.

Client conversation:

When presenting campaign impact to a client, reporting lift as a range rather than a single number can land better with senior stakeholders. A statement like "+12 % ± 3 pp" is more honest than a bare "+12 %" because it acknowledges that measurement carries uncertainty. Framing the credible interval as a planning range gives a media planner two concrete scenarios to work with, which can be more actionable than a point estimate.

The method choice itself can also do work in the conversation. Being able to say "we used BSDID because it protects against a specific type of measurement error that the standard method misses" gives the client a concrete reason to trust the number, and the per-week posterior plot is a natural companion when walking through the results, showing whether the effect ramped up, stayed flat, or faded rather than reducing the whole campaign to a single headline.

Finally, an honest report that acknowledges uncertainty naturally opens the next conversation. When a client asks "how do we tighten the range on the next campaign?" it becomes an easy bridge to discussing a bigger panel, a longer pre-period, or priors drawn from earlier campaigns. In procurement or analytics-heavy conversations, being able to point to the robustness check across 20 simulated scenarios is a straightforward answer to "how do you know this method works?".

8 Limitations

This is a controlled method-comparison experiment where the ground truth must be known exactly, so several simplifications are deliberate: constant treatment effect, iid Gaussian noise, homogeneous control regions, and a generative structure close to what the models assume. The synthetic setup is cleaner than real geo data, which has non-stationarity, structural breaks, autocorrelated noise, regional shocks, and heteroskedasticity. Absolute numbers (BSDID 19 % lower mean absolute bias, 9 % tighter interval on the 20-seed Monte Carlo) reflect this synthetic setup and would move on real data.

A 20-seed Monte Carlo study is included. Empirical coverage at the nominal 90 % level is 75 % for both models, indicating slight residual overconfidence. A production-grade implementation would add informative priors on omega and lambda (e.g., tied to geographic similarity or historical experiments), hierarchical pooling across past lift tests, and a larger Monte Carlo (100+ seeds) to sharpen the coverage estimate. Priors are currently uniform Dirichlet on both weight simplices and treatment is pure on/off.

BSDID's theoretical advantage over BSC is that it subtracts a systematic pre-period offset between treated and synthetic control before computing the ATT. That advantage is most pronounced when the treated unit has a differential trend, structural drift, or persistent offset that BSC would misread as post-period treatment effect. The synthetic panel here has homogeneous trends across units and only a small pre-period offset on the treated unit, so BSDID's correction has relatively little to correct. On real geo data with differential regional trends and macro drift, BSDID's margin over BSC is expected to widen.